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Water optimization technique for precision irrigation system using IoT and machine learning

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ABSTRACT

Water management system and energy optimization are most important for today's agriculture. Precision irrigation manages the water utilization and reduces the utilization of energy. The irrigation system consists of observation system, storing, processing and it helps in decision making as well. Internet of Things (IoT) components is used to observe the various information's, such as soil moisture, soil temperature, weather conditions, and environmental conditions from the fields. With the help of various information, the proposed work optimizes the requirements and thus reduces energy. The cloud environment is used to store the observed parameters from the IoT components. The machine learning algorithm is useful in taking the process of data, from the cloud environments, and do forecasting on the irrigation system. In this work, proposed a hybrid model for irrigation system and do the decision making using, different observation of parameters. This work consists of K-Nearest Neighbors (KNN), Gradient Boosting-based Tree (GBT), Long Short-Term Memory (LSTM) and Spearman's rank correlation. The KNN is used to gather the nearest sensing information, with the help of various parameters. The GBT is used to predict the real values based on the observations. The LSTM is used to predict, time series prediction using different observations of the surrounding. The Spearman's rank correlation is used to receive the correlated values of real, and time series prediction. Using the GBT, LSTM and spearman rank correlation, the consolidated time series forecasting values are predicted in irrigation. The implementation of proposed work is done with the help of banana species in the time period, between Feb 2020, Dec 2020 and 2021. The duration of the cultivation, minimum and maximum water requirements and other supporting parameters are calculated using various sensors. The real and time series water requirements are calculated in the time interval of 12, 24, 36 and 48 etc. With the help of this time series data, month-wise prediction is performed. The implementations were performed in single and group of banana tree water requirements. After implementation of proposed work, for single banana tree 31.4 % of the water requirement has been optimized in 2020 time period. Due to this optimized irrigation, heavy usage of fresh water usage and energy wastage is also reduced.

Introduction

Agriculture is the biggest field which consumes fresh water and electricity in the whole world [1,2]. About 70% of water is used for agriculture and it consumes huge units of electricity, as well. Out of this, only around 25% of water and electricity are being consumed efficiently, and the remaining is being wasted away [3,4]. Water is useful for entire process of agriculture and electricity is being used in various activities such as planting, sowing, spraying, weeding, and harvesting. So, the usage of fresh water and electricity is high, likewise, the wastage is also high. The optimization techniques reduce the usage of water and electricity [1,2]. On the other hand, the effective usage of

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water increases crop productivity. Thus, for the efficient usage of water and electricity, optimization method is needed for today's society.

The trending technologies have decreased the water and electricity usage in various fields and it is essential in today world. Different electronic components, techniques, algorithms have been introduced by different researchers on optimization [4]. Nowadays, IoT and machine learning technologies, helps in many ways to decrease the energy usage and to increase the productivity in many fields. Few application fields of IoT and machine learning are, Industries, Road Transport, agriculture, Optimizations, etc. This work has used both IoT and machine learning for energy optimization and effective utilization of water in irrigation systems. IoT is the combination of interrelated devices such as firmware, mechanical, and few programming as well. The IoT devices are used to collect data, and, the collected data are being managed, using cloud computing. Machine learning is an automated technique that improves efficiency through past experience. Some of the efficient machine learning algorithms and mathematical models are Bayesian Statistics, KNN, Neural Network, Support Vector Machines (SVM), Decision Tree (DT), etc. These learning mathematical models are used to make effective analysis and decision-making in various applications.

An article [27] has been introduced to automate the agriculture process as small, moderate and high to reduce the cost. KNN, Logistic Regression (LR) and DT algorithms have been proposed to handle loads of agriculture for 562 former cases. An advanced machine learning based agriculture for smart cultivation has been proposed [28]. In this work, parameters like phosphorus, rainfall, surface temperature, crop rotation have been taken to test performance of smart agriculture. Multiple linear regression was the algorithm introduced to check the performance.

The IoT and Machine learning-based technique for energy optimization and effective utilization of irrigation systems has three processing layers, such as:

i. Data gathering/collection and transmission
ii. Data processing and intelligence layer
iii. Application Layers

The above layers have different functionalities and components. Data gathering and transmission are performed in the first stage and it collects the information from the fields or agricultural grounds, and transmits for data processing. For the data collection and processing, certain physical components (sensors) are used. The components of data collection and transmissions devices are, Wi-Fi or Mobile data connections, ZigBee Network for local area data processing, Wireless sensor Node (Single Sensor Node, Multiple sensors Node) for data collection from the ground and atmosphere, Raspberry Pi, controlling components and other physical devices, are for controlling the motors [5]. The second layer is data processing and intelligence. It is used to process the gathered data with the help of machine learning and intelligent techniques. Using the second layer, decision-making is performed. The third layer is the application layer; it receives the data from the second layer. Based on the second layer's output, the third layer is planned for irrigations and optimizations [7]. Entire data is stored in the cloud for future planning. The layers, components of IoT, and machine learning algorithm for general irrigation system representations are shown in Fig. 1.

Fig. 1 mention about each layer, and overall structure of data processing, which has different challenges. The main challenges which are related to layers and overall structure are as follows.

i. The entire irrigation applications are not fully automated.
ii. The integration part of each layer has different difficulties such as data collection from the bottom layer to transmitting to the next layers.
- The integration of different processing layers.

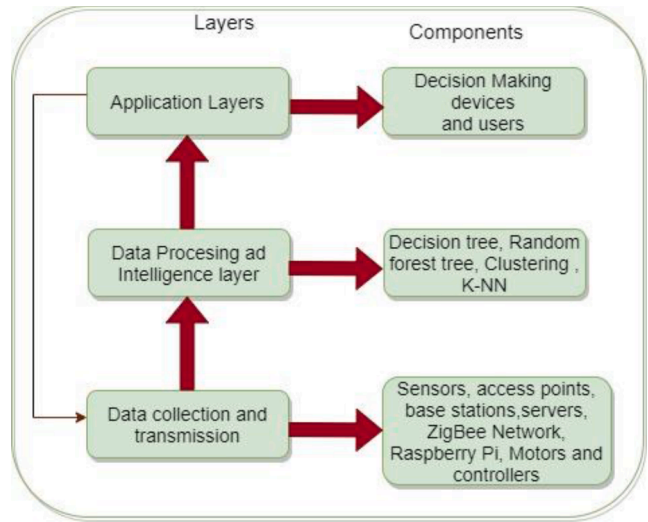

Fig. 1. Layers and Components of irrigation system.

- Land slope measurement calculation and water calculation of each entire land.
- Water optimization is calculated based on the rain, moisture, and atmosphere activity.

These are the main issues of the irrigation system. This proposed work, considers the following issues, such as moister level of ground and soil of plants, land slope monitoring, and maximum utilization of plants, month-wise water usage, optimized usage based on power utilization.

The topics discussed in this research paper in the subsequent sections are as follows. Section 2 discusses current and related work on irrigation methods, various projects carried for irrigations. Section 3 delves into the working methodologies, components based on IoT and machine learning. Section 4 contains the implementation and performance analysis of the proposed work along with the existing work, followed by the conclusion.

Related works

The supporting work of this article has two parts. One is, smart irrigation techniques, frameworks, and models for agriculture. The second one is, Machine learning techniques and algorithms for irrigation system.

Irrigation techniques and models

Many researchers have presented, different articles, related to the smart irrigation system with IoT, cloud, and machine learning [3,7]. The researchers have focused on how high technology is incorporated to effective smart irrigation, for data transmission and decision making. This related work is based on IoT data processing, centralized data storage for decision making, and machine learning techniques for decision making. The researchers of [3] have presented an IoT framework for data processing, storing in the cloud, and analysing data from, the processing devices with a decision support system. The authors of [4] have presented an IoT platform reference model support for intelligence IoT agriculture. The authors of [5] have presented an IoT Intelligence application in landscape and many dimensions. The main supporting dimensions are land slope, moisture of soil, water evaporation is the main dimension which is considered in this article.

Many researchers have considered different parameters for intelligence irrigation systems. Authors of [6] have presented a climatology and geography model, with important factors such as moisture of soil, the wetness of the soil, monthly and daily soil moisture, weather report,

evaporation of the soil are considered for smart irrigations. The Crop Water Stress Index (CWSI) model is presented in [7] framework to achieve water saving, and irrigation management, based on shade temperature distribution of plants. In this work, water is optimized in the interval of time and continuously checked on the plants. The authors of [8] have presented an automatic irrigation technique to schedule the direct water soil management and with this technique, the efficiency of water utilization is improved. The evaporation-based techniques [9,10] have presented a control-based scheduling system for irrigation. The main advantage of this method are the different parameters, which are considered here, such as, wind, velocity, humidity, and soil moisture, etc. The authors of [11] have presented a sensor-based irrigation system. In this system, sensors, sense the soil moisture, and the wet and dry status of soil is also considered. Mobile application is used to monitor, various activities of irrigations. The recommended system of irrigation system [12] presents various activities based on machine learning. In this work, regression model, Deep Neural Network (DNN) and GBT methods are used to increase the prediction. The accuracy of the prediction and recommendation are improved, and the accuracy rate of prediction, in various parameters is 93%. The researchers of [13] have presented an intelligence monitoring system using thermal images. The imaging camera is mounted on the drone and captures image based on various parameters, such as water requirement, leaf potential, non-uniform irrigation, etc. The authors of [4] have presented IoT and machine learning-based smart irrigation systems. Using this system, various parameters are calculated such as environmental parameters, various soil conditions, air temperature, and nearest futures are considered. The [14] have presented an optimized watering system for crops, using Wireless Sensor Network (WSN). The main aim of this work is node sensors, which are used to control and manage the crop-based web application and mobile applications. The soil moistures are calculated and data are managed. Using this data, future information is calculated and predicted. The water management system using IoT with multi-sensor [15] presents a new technology with inexpensive efficient component, which is used to measure different parameters such as temperature, humidity, soil moisture, and rain sensors. The predicted output is controlled manually and automatically. The authors of [19] presented various monitoring and control system for irrigation system. The authors of [20-23] proposed an improved energy efficiency model using IoT platform. Using this platform, parameters and pumping operations decision are performed. The authors of] proposed a model for energy efficiency and smart control to cover 5 km. In this model the information's are transferred using LoRa network [24-26]. The authors [29-33] described the how reinforcement learning is used for irrigation system and how to optimize the water in irrigation system. The summary of different models, advantages and limitations are represented in Table 1.

Table 1
Various Irrigation Techniques along with merits and limitations.

works	advantages	Limitations
IoT framework for Irrigation [3], 2013.	Data is processed, stored, analysed and decision are performed using IoT	Requires to considered parameters for calculation.
IoT platform model [4], 2018.	Supports intelligence decision making.	
IoT Intelligence application for irrigation [5], 2016.	Supports intelligence irrigation system with various supporting factors.	Requires different parameters such as length and breadth of root moisture.
Climatology and geography model for irrigation [6], 2007.	Various factors are considered for smart irrigation system	Requires the length and breadth of root moisture, land slope etc.
CWSI Model [7], 2010.	Achieves water saving and continuously checks the moisture of plants.	Requires data storage and optimization.
Automatic irrigation technique [8], 2010.	Automatic scheduling techniques for irrigation and improved water utilization.	Requires to consider different parameters to improve the optimizations.
Evaporation-based techniques [9,10], 2017.	Control based scheduling for irrigation with various parameters and crop conditions	Requires considering data storage, operations and soil parameters.
Sensor based Irrigation system [11], 2015	Soil moisture, dry and wet status of land is calculated with the help of mobile applications.	Requires centralized storage, prediction water usage and planning.
Recommendation system of irrigation system [12], 2017.	Using machine learning, it increases recommendation and prediction.	Requires improving the accuracy of prediction and recommendations with various parameters.
Intelligence irrigation monitoring system [13], 2017.	With the help of drone, various parameters of irrigation's are calculated.	The accuracy and predictions based on drone is difficult to monitor.
IoT and machine learning based smart irrigation system [4], 2018.	Using IoT and machine learning various parameters are considered for effective decision making and smart irrigation.	Requires considering multiple node-based calculation and land slope is also needed for water utilization and calculation.
IoT and data analysis for smart farm [14], 2019.	Node sensors are used to control and manage the crop, based on web application and mobile applications. Using this, the soil moisture is also managed and future data is also predicted.	Requires considering many different parameters for prediction of managing data.
Water management system using IoT with multi-sensor [15], 2019.	Inexpensive components are used to manage different parameters. Produces optimal solution to smart agriculture.	Requires considering depth of the soil moisture and environmental parameters.

Machine learning techniques and algorithm for irrigation

In agriculture, for data prediction, and analysis, different types of machine learning algorithms were used. With the help of machine learning, environmental data, irrigation management, moisture analysis are predicted and recommended for the future. The work [1] introduces irrigation planning using a genetic algorithm. Using this work, maximize the output, and calculate various recommendations made for irrigations. The authors of [2] provide solutions for sequential irrigation issues with the help of a genetic algorithm. The proposed genetic model provides a solution using non-continuous scheduling. The researchers of [3] proposed a weekly irrigation plan with supporting data such as weather and metrological data. The boosted regression and classifier, which are used for data analysis, is based on this irrigation recommendation. With the help of this, 93% and 95 % accuracy are achieved in regression and classifications. The optimized irrigation allocation method, is introduced in [4] for location-wise demand of water. The genetic algorithm, which is introduced for optimizations, uses the genetic algorithm with previous day's data, so, new irrigation plan is introduced. The Intelligence IoT with [5] KNN machine learning method is used to analyse the sensor data for planning the irrigation. This work is fully automated, and data is communicated from machine to machine. The work of [6] machine learning for farm enhancement, was introduced for heterogeneous data using IoT. In this work, decision trees are used to predict the related data for irrigation. The authors of [7] uses, logical regression to predict the water requirements from time to time. The works of [8,9] decision tree, is use to predict the speed of the wind, temperature, and humidity. The SysFor [10] provides better results compared to the other decision trees and provides better accuracy in prediction. The irrigation demand prediction SVM [11] algorithm is used, but the accuracy of recommendation is very less.

The related works and existing works of irrigations methods have various limitations. Some of the limitations are as follows.

- The most popular works do not consider the land slope and water flow in the agricultural ground.
- Different machine learning algorithms are used for prediction. But most of the works has only one classification or prediction algorithm which is used for moisture calculation, wind, and humidity calculation. So, the predictions do not provide accurate results.
- Most of the works and solutions are manual-based. Some of the smart devices produce automation.

Contributions: The main contributions and novelty of this work is,

i. optimize the water and save the energy in banana agriculture using IoT devices and Machine learning algorithms.
ii. Especially, the optimized technique is applied into banana varieties and maximum water and month wise usage of water is fully calculated and to indirectly reduces the energy usage.
iii. Compared to the traditional irrigation system in banana agriculture, remarkable percentage of water is optimized using this proposed smart hybrid irrigation system.

Materials and methods

This work was carried out in department of Computer Science Engineering, Galgotias University, India. Four different algorithms have been taken to yield water optimization. The data is collected from real time Nendran variety banana cultivation in some places like Kanyakumari and Trichirapally of India with different epochs ranging from 12 to 72 for dataset collected from February 2020 to November 2021. The pretest power (G Power) obtained is 80 % and alpha value is 00.5. The dataset training and testing calculation of various durations is listed in Table in section 4. The outcome of this proposed hybrid method provides an optimized solution for irrigation, energy, and increasing the yield of crops. Using the optimization reduces the cost of agriculture and increases productivity. In this proposed method, 3 algorithms have been connected and correlated and hence, the proposed idea would produce better performance. KNN machine learning algorithm has been used to sensing information of agriculture land and neighbors land information. To predict observation based real field results, GBT has been introduced appropriately. Finally the LSTM algorithm was used for ranking correlation among gained results. For water requirement parameters, KNN, GBT and LSTM algorithms have been used to predict the efficiency. Further Spearman's rank correlation has been called out to make decisions. Thus this proposed idea introduces novelty for the dataset and thus novelty is manageable as the dataset grows. The proposed system and the layers are shown in Fig. 2. The proposed system has three layers.

i. data collection and transmission layer: It is used to collect and transfer the data from the ground level. This layer is used in different components, to collect and transmit the data.
ii. Data processing and intelligence: it is used to process the data from the first layer, and intelligence techniques are used to make decisions. In this layer, machine learning techniques are used to make the decision, and based on the decision; the irrigation planning is being done.
iii. The application layer is used to visualize the data from the data collection and processing layers. The application layer continuously monitors the present, previous and historical data of the crop.

Data collection and transmission layer

The Data Collection and transmission layers are used to collect and store data in cloud environments. The main component and their corresponding usage of each component are the works shown in Table 2 and Fig. 3.

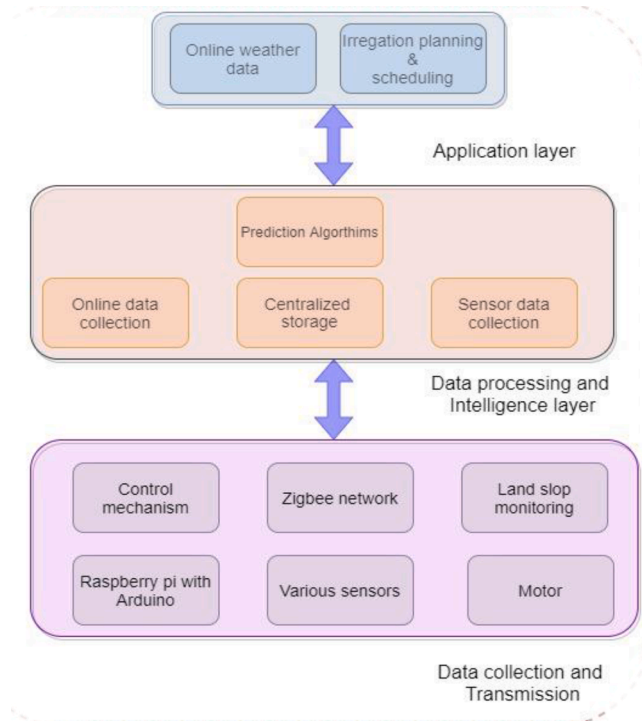

Fig. 2. The Proposed Model Architecture and Layers.

Table 2
Various Components in smart irrigation system.

S. No	Components	Usages
1	Digital Inclinometer	Used to measure the slope gradient of land structure and excavation slope.
2	soil moisture sensor –F-28	Used to measure the moisture of the soil.
3	Water level Monitoring sensors (HPT675 and RS485)	Used to measure the level of water.
4	Temperature Sensor (THERM200)	Measure the soil temperature
5	Humidity Sensors (HTM2500LF)	Measure the humidity and Water vapour
6	Root Depth Moisture (SHT11)	Used to measure the moisture and depth of the soil.
7	WSN Environments	Surveillance, Monitoring and transmission
8	ZigBee Network	High Level Network which is used for personal Communications.

Various sensors are used to collect data, in WSN environments. The WSN environments process the data internally and transfer it to the cloud environments. Various sensors and Digital Inclinometer are used to monitor the land and status of soil.

The digital inclinometer is used to find the pattern of the slope and deformation in the study state. With the help of a digital inclinometer, it can easily verify the stability of the slope and give warning on the status. The data collections using various sensors are represented using Fig. 4.

Data processing and intelligence

The data processing and intelligence consists of proposed intelligence methods to process the data and make decision. This proposed method is performed with the help of an ensemble-based hybrid model. The hybrid model consists of KNN, LSTM, and GBT. The KNN is used to find the nearest values from the root of the plant. GBT is used to find both regression and classification. With the help of GBT, it can improve the real prediction accuracy with sufficient optimized value. The LSTM is used to predict the time series prediction value and with the help of

Fig. 3. Component's representation of proposed work.

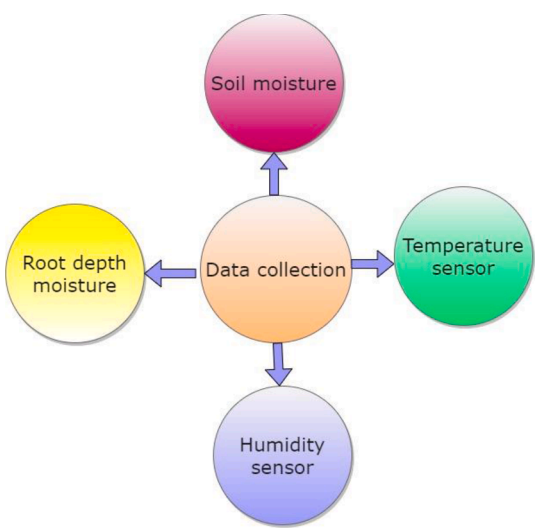

Fig. 4. Data Collection from various sources.

time interval data it can be predicted.

K-Nearest neighbours Algorithm

The KNN algorithm [16] is used to find the nearest values from the predicted sensors. It is a non-parametric algorithm, and without any assumption, it produces the real values. The Input consists of K nearest values in the dataset or data, from the sources. The output of the prediction depends on the nearest neighbour's distance or prediction. The output class is predicted using the vote of its nearest prediction. The training data of KNN has multiple multi-dimensional features and space in each class of the supervised label data. The prediction values depend upon the continuous or discrete distance of the features. The distance between the feature's predictions depends on Euclidean Distance. The representation of nearest neighbours' prediction with various × parameters is shown in equation (1).

$$Y = C_n^w(x_1, x_2, \cdots, x_n) \tag{1}$$

Y - is the predicted output, *C* - is the nearest features, *n* is the number of features, *w* is the weight, $\{x_1,\ldots,x_n\}$ is the various weight. The distance between each feature × is defined using the Euclidean equation is shown in equation (2).

$$D = \sqrt{\sum_{i=0}^{n}(\mathbf{x_i} - x_j)} \tag{2}$$

D - is the distance, *n*- is the number points of particular features ×, i - and j - are the starting and ending points.

Gradient boosted tree

The GBT regression is the machine learning algorithm for classification and prediction which generates a real prediction tree [17]. The gradient boosting is a decision tree, to a base learner. The gradient boosting combines multiple bases, which are, weak leaner and strong leaner, in the iteration process. The GBT has *m*-th step decision tree denoted as *dm(x),* to pseudo-residual. The Nm is the number of leaves. The decision tree is portioned into Nm, which has disjoint space $\{S_{m1}, S_{m2},\ldots,S_{Nm}\}$ and prediction values of each space. The output of *dm(x)* for various input features (x) is represented in equation (3).

$$\mathbf{d_x}(x) = \sum_{1}^{N_m} \mathbf{b_m S_{1m}}(\mathbf{x}) \tag{3}$$

where, b_{mi} are the predicted values of each space of S_{im}. This GBT model builds the stage-wise fashion and builds the optimized model in each stage. The various outputs such as moister, temperature, and humidity is observed at every 12 h of prediction. Then, the ARIMA model is used to generate time serial analysis of the various parameters in the time series. Using this model predict the future of various parameters using current and future data. The representation of the ARIMA model is shown in equation (4).

$$\left(1 - \sum_{i=1}^{p'} \alpha_i L_i\right) X_t = \left(1 + \sum_{i=1}^{q} \theta_i L_i\right) \varepsilon_t \tag{4}$$

where L is the operator, ∞ is the parameters of autoregressive, ψ is changing parameters, θ is error rates, X_t is time-series data. The input data is generated using various time intervals such as 12, 24, 36 and 48 h and based on that a real data requirement is predicted.

Long short-term memory:

The LSTM is an artificial recurrent neural network and it works on deep learning, it is used to classify and predict the data based on the time series. The LSTM consists of an Input gate, forget gate, and output gate. The cell remembers the time interval, and the three gates are used to control the flow of collected information with the help of entering and exit gates. The function of forget gate is represented in equation (5).

$$f_t = \sigma_q(w_f x_t + u_f h_{t-1} + b_t \tag{5}$$

where, f_t –forget Activation function, h_{t-1}– hidden vector of previous unit *t-1*, x_t – input unit, b_f – bias parameter,σ_q sigmoid function. The sigmoid function values are generated between 0 and 1. 1 represents information is in remember and 0 represents that the information is forgotten. The sigmoid function is represented as *S(t)* in the equation.

$$S(t) = \frac{1}{1 + e^t} \tag{6}$$

The function for which new information is added in remember cell, is represented in equation.

$$i_t = \sigma(w_i(h_{t-1,} x_t) + b_t \tag{7}$$

The information which needs to add to the new state (S_n) is shown as.

$$S_n = f_t * S_{t-1} + t_t \tag{8}$$

Using this LSTM, the past 4 to 6 data has been collected simultaneously in the interval of 12 h. Various supportive data such as moisture, speed of wind, and direction are collected at 12 h interval.

Hybrid model

The structure of the proposed hybrid model flow representation is shown in Fig. 5. The structure of the proposed model consists of KNN, GBT, LSTM, and Spearman's rank correlation. Using this correlation, the co-efficient, without assumption predicts the values at various time intervals. The model is designed with the same configuration in three different places for data collection and processing. So, the model of linear, with non-parametric parameters is collected frequency in the different distribution. The non-parametric values are calculated by correlation co-efficient in 12 h' time interval, every day. The data is collected at 12 PM and 12 because during FN the moisture range is changed according to different climatic conditions, so easily it can predict the real and requested data. The structure of the proposed workflow consists of three types of models that predict the non-parametric values at the same time and is ranked using Spearman's rank correlation (SRC). The observed values are ranked using co-efficient and correlation functions.

In the proposed model algorithm 1 KNN_n, GBT_n, and $LSTM_n$ are the forecasted values of KNN, GBT, and LSTM. The n is the number of inputs and R_n is the observed values. From the second step hours is initiated and is predicted by hours loop. The correlation coefficient of models such as KNN, GBT, and LSTM is calculated in terms of reality, forecast, and correlation co-efficient of the best model, and then it creates the best liner model for predicted values and observed values. The R (hr) and I (hr) such as regression and interception are calculated simultaneously. Finally, the best values are replaced by the proposed hybrid model within 12 h' time interval.

Algorithm 1: Hybrid Model.
Input: KNN, GBT, LSTM, R_n, and various parameters (n).
Output: Hybrid forecasted values.

1. **Begin**
2. **Interval ← 1**
3. **While (Interval ≤ 12) do**
4. $KNN_{SR}^{n} \leftarrow$ **Person** (KNN_{SR}^{n}, R_n)
5. $GBT_{SR}^{n} \leftarrow$ **Person** (GBT_{n}^{hr}, R_n)
6. $LSTM_{SR}^{n} \leftarrow$ **Person** $(LSTM_{n}^{hr}, R_n)$
7. $R_{SR} \leftarrow \{ KNN_{SR}^{n}, LSTM_{n}^{hr}, GBT_{n}^{hr} R_n \}$
8. $Best_{SR}^{h} \leftarrow$ **Max of** $\{ KNN_{SR}^{n}, LSTM_{n}^{hr}, GBT_{n}^{hr} R_n \}$
9. $R(hr), I(hr) \leftarrow \{ Best_{SR}^{h}, R_{SR} \}$
10. $hr \leftarrow hr + 1$
11. **While (hr ≤ 12) do**
12. $Hybrid_{n}^{hr} \leftarrow R(hr)^{*}I(hr)^{*} Best_{SR}^{h}$
13. **End**

There is a relationship between machine learning algorithms KNN and GBT with Spearman's rank correlation since predicted value is correlated with real value of water irrigation.

Results

In the experimental and data analysis, the Banana (Musa) crop irrigation data is being used. Different varieties of banana specifications are represented by the national horticulture board of India [18]. In this work, Nendran variety cultivation fields are used to collect the data, because it requires large amount of water. Data is being collected from the southern part of India, Kanyakumari (8.2604° N, 77.2671° E). Two major reasons for choosing this location. i. Nature of soil, which supports different types of crop culture, and the climate condition, also changes dynamically, each and every month. ii. In this location, different types of banana crops are cultivated. Basic data requirements are collected with the help of farmers and the National Research Centre for Banana, Tiruchirappalli, India. Basic water requirements, number of nodes and duration of cultivation, and other parameters for Nendran variety are shown in Table 3.

Various parameters of observations, from the different sources, are shown in Table 4. In this table, different parameters are considered to evaluate the requirements of water calculation of concerned crop. Maximum water requirement per month, moisture of the surrounding, root moisture, humidity rates, rates of rain, maximum water

Table 3
The basic Requirement of Nendran Banana (Musa).

Species Name	Banana Crop
No of Nodes/ Tree	1000/ Per Acre
Duration	9/10 Months [Feb- Oct/Nov]
Maximum Water Required	360/400 Litres
Starting date of Agriculture	Feb-01, 2021
Max Water Requirements of each month	50 Litres
Interval of irrigation	7 Days

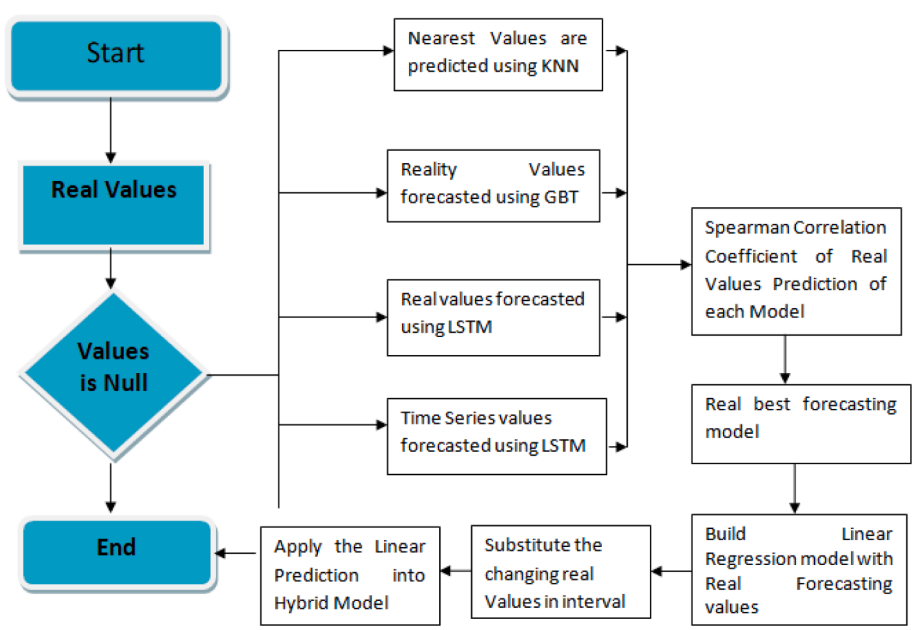

Fig. 5. Hybrid Model flow Representation.

Table 4
Summary of various parameters and water requirements.

Months/Parameters	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov
maximum Water	40	40	40	40	40	40	40	40	40	30
Moisture surrounding	0.56	0.52	0.78	0.8	0.9	0.89	0.83	0.85	0.78	70
Root Moisture	0.88	0.8	0.78	0.82	0.98	0.97	0.91	0.88	0.8	0.78
Raining	0	0	0	0	5.0	10.2	12	12.2	13.0	8
Humidity (Max)	32	33	31	30	25	24	24	30	31	30

requirements, calculated using various parameters, are also shown in Table 4. Finally, required water is also calculated, using, the above-mentioned parameters.

Data pre-processing consists of data collection, training, testing, and validation. The data is collected in 12 h' time intervals using LSTM. The data is collected from two zones of cultivation fields and of two cultures. The requirements and other data are collected from Feb 2020 to Nov 2020 and Feb 2021 to Nov 2021. In Table 5 two cultivation training, testing, and validation data are given in different time intervals. The dataset is used for supervised learning patterns and the parameters were normalized on the same scale. The min–max scalar is used to standardize the dataset, and the corresponding equation, and data splitation is also shown in Table 5. The min–max scale is performed using equation no.

$$Scaled = Std^{*}\ (Max - Min) + Min \tag{9}$$

where, *Min, Max* is the feature, which ranges between the minimum and maximum values. The coefficient of determination (R^2) and root-mean-square error (RMSE) are used to predict the variation and value observations.

Discussion

This section may be divided by subheadings. It should provide a concise and precise description of the experimental results, their interpretation, as well as the experimental conclusions that can be drawn.

In this work, each epoch, has the best model data, which is calculated from the two different cultivation fields, and best-optimized data is observed, using, Spearman correlation. The variance and observation of the best consolidated data model are calculated using R^2 and RMSE. In this hybrid model, the KNN algorithm is used to predict the nearest parameter values and the other two models GBT and LSTM are used to predict the requirements of water. The epochs are calculated based on the 12, 24, 36, and 48-hours intervals. The coefficient of GBT and Spearman correlation in 12 and 24 h is 0.72 and 0.78 which is comparatively more as R^2 was achieved as 0.70 [7]. The LSTM spearman correlation is 0.68 to 0.72. The hybrid model of zone 1 and zone 2 values are calculated and consolidated in 12 h time intervals. The R^2 values and RMSE values were also predicted in 12 h duration and the consolidated values are shown in Table 6. The R^2 values of GBT and LSTM ranges between 0.93 and 0.99 and 0.92 and 0.99 and once again more almost 20 to 20 percentage [7]. Therefore, no major difference between GBT and LSTM values in terms of variance in zone 1 and zone 2. Similarly, the RMSE values of GBT and LSTM are 0.23 to 0.7 and 0.22 to 0.6 in prediction in zone 1 and 2 when compare with existing work, RMSE is efficient [7]. The consolidated values of the prediction of various epochs are shown in Table 6. The combination LSTM and GBT is similar in performance to zone 1 and 2, during different time intervals. Similarly, the performance time (PT) of LSTM and GBT ranges of zone 1 and 2 are 12 s to 39 s per epoch [8,9]. Finally the Spearman correlation is decreasing with number of epochs increasing where as the R^2, RMSE and PT were increasing with number of epochs increasing.

The irrigation water prediction using GBT, LSTM and hybrid models in various time intervals such as 12, 24, 36 and 48 for zone 1 and zone 2 are shown in Fig. 6 (A-C).

Fig. 6 data change dynamically as per the changes of Table 4 parameters. Based on the Fig. 6 (C) hybrid prediction, the month-wise water requirements and water optimization are shown in Figs. 7 and 8.

The water requirement and water optimization of 2020 and 2021 calculated with the help of minimum requirements (40 Liters) of water for cultivation. In the starting of the cultivation (Feb 2020 and 2021) the water requirements is very less because the root length of banana tree is very less in initial stages. But after one month the water requirements is increased automatically. The second and third month wise the root of the banana tree is increased and spreads surrounding. So automatically the requirements of water increased up to 6 to 7 months. The last two month of the cultivation again the requirements of water reduced, because the nip of the root is decomposing, so automatically again water observation is reduced. In the prediction of water requirements of two cultivations, as per Fig. 6 (A) and Fig. 7(B) requirements the rain fall is very less in 2020. So water requirements of tree are increased. But in 2021, initially the rain fall is less but last three months the rainfall is high in cultivated location. So the water requirements are 0 in last three month of cultivation in 2021. The entire water requirements and optimized water of single tree banana cultivation (2020 and 2021) are shown in Table 8. The 2020 single tree requirement of water is 273 and optimized water of single tree is 127, so 31.14 percentage of water is

Table 5
Training, testing and validation ratios of Water Requirements.

Hours	Zone	Water Requirements	Training Data (60%)	Testing Data (20%)	Validation Data (20%)	Total
12	1	Min	3900	1300	1300	6500
		Max	3120	1040	1040	5200
	2	Min	3480	1160	1160	5800
		Max	3120	1040	1040	5200
24	1	Min	2880	960	960	4800
		Max	2520	840	840	4200
	2	Min	1920	640	640	3200
		Max	1680	560	560	2800
36	1	Min	1200	400	400	2000
		Max	1080	360	360	1800
	2	Min	900	300	300	1500
		Max	720	240	240	1200
48	1	Min	660	220	220	1100
		Max	540	180	180	900
	2	Min	480	160	160	800
		Max	360	120	120	600

Table 6
Spearman correlation, R^2, RMSE and performance time values of GBT and LSTM.

Zone-1 GBT epochs	S.C	R^2	RMSE	PT(s)	Zone-2 GBT S.C	R^2	RMSE	PT(s)	Zone-1 LSTM S.C	R^2	RMSE	PT(s)	Zone-2 LSTM S.C	R^2	RMSE	PT(s)
12	0.723	0.94	0.437	15	0.73	0.95	0.357	14	0.68	0.93	0.537	25	0.71	0.92	0.237	18
24	0.732	0.95	0.347	13	0.75	0.93	0.247	13	0.71	0.92	0.347	13	0.68	0.96	0.247	16
36	0.716	0.93	0.432	24	0.72	0.94	0.532	28	0.71	0.93	0.332	18	0.69	0.93	0.532	17
48	0.713	0.97	0.542	36	0.78	0.96	0.542	26	0.72	0.95	0.442	26	0.72	0.94	0.542	21
60	0.724	0.96	0.712	35	0.74	0.95	0.612	27	0.72	0.99	0.512	27	0.72	0.97	0.612	26
72	0.681	0.99	0.452	36	0.68	0.99	0.552	39	0.71	0.97	0.652	26	0.71	0.99	0.552	37

Note: S.C - spearman correlation, R^2 - Coefficient of Determination, RMSE - Root-Mean-Square Error.

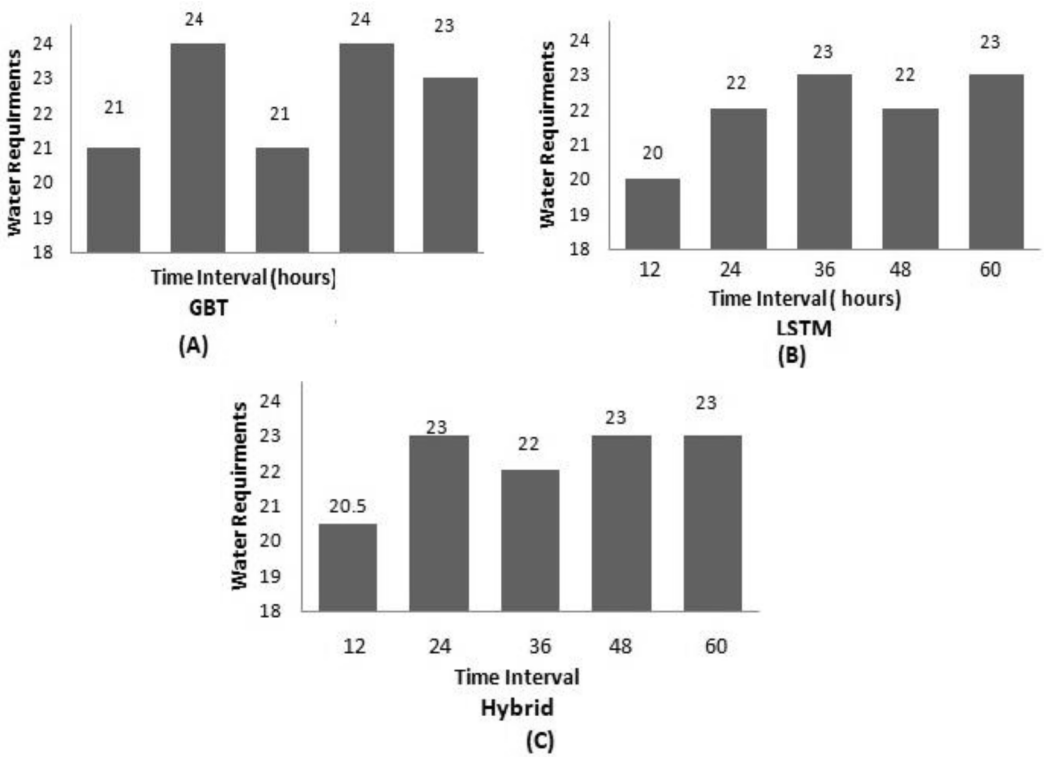

Fig. 6. Prediction of water Irrigation using GBT, LSTM and Hybrid models using various time interval.

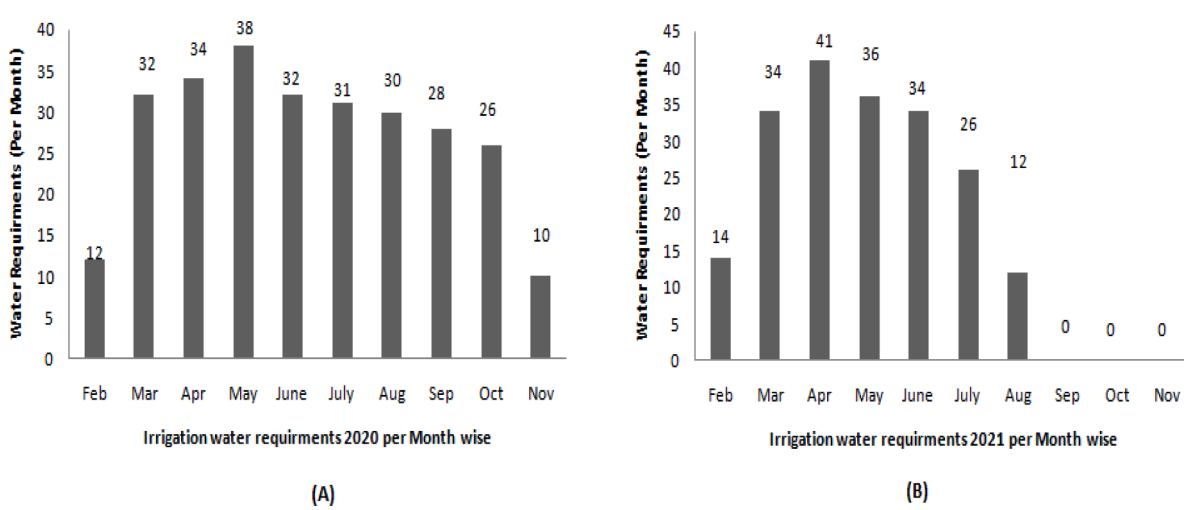

Fig. 7. Month wise water requirements in 2020 and 2021 Cultivation.

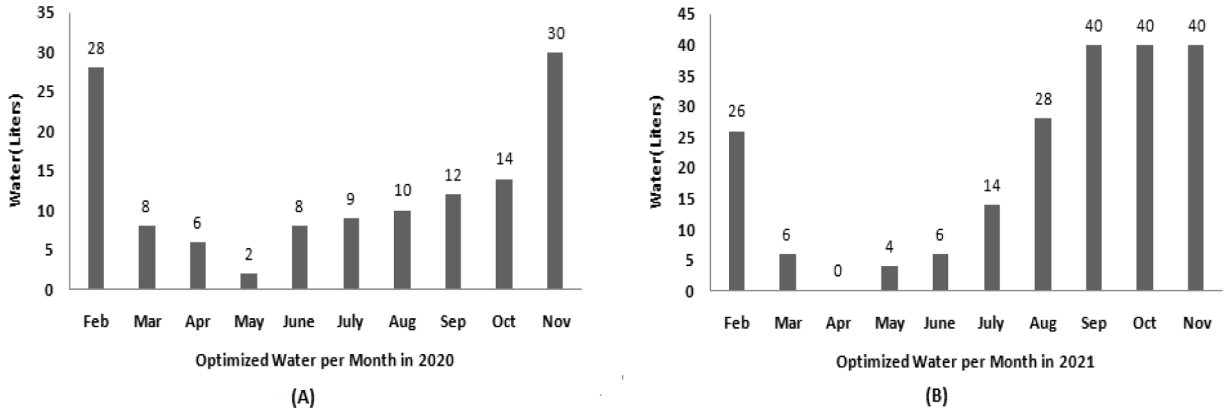

Fig. 8. Month-wise Water optimization in 2020 and 2021.

Table 8
Water requirement and optimization of 2020 and 2021 cultivation of banana.

Water requirement /Optimization	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov
Water Requirement 2020	12	32	34	38	32	31	30	28	26	10
Optimized Water in 2020	28	8	6	2	8	9	10	12	14	30
Water Requirement 2021	14	34	40	36	34	26	12	0	0	0
Optimized Water in 2021	26	6	0	4	6	14	28	40	40	40

optimized for single tree [15]. Similarly, 2021 due to heavy rain in that cultivation around 50.1 percentage of water is optimized [20]. Due to this water optimization, indirectly the energy usage also reduced.

The results of all the parameters such R2, RMSE and PT were compared with existing works [71520] for performance evaluation. It is very clear that R2 is best with the proposed approach with considerable gain. The work [7] claimed more RMSE with agriculture cultivation with IOT and thus proposed work is best suited dynamically. Finally work [20] measured the PT of agriculture time and it is compared with proposed work and claimed considerably less time than existing work. Spearman's rank is used to correlate the real and time series values of water irrigation after receiving predictions. In Kanyakumari (8.2604° N, 77.2671° E), two weather based climate seasons are introduced, but if only one climate based location for which this proposed solution is proposed then certainly some changes have to be made to reduce water usage and thereby increase in efficiency.

Conclusion and future work

In this work, proposed hybrid model used to predict and forecast the requirement of irrigation using various parameters. The proposed method has built using KNN, GBT, LSTM and Spearman's rank correlation algorithms. The KNN is used to collect different parameters with the help of sensors. The GBT is used to predict the real requirements of water at the specific time and the LSTM is used to predict the time series wise predict the requirements in the different time intervals. Using Spearman's rank correlation the predicted and forecasted data is consolidated and sent to decision making. The combination of GBT and LSTM the real time series data is forecasted as various time intervals. With the help of the forecasted data, a month wise and entire cultivation requirement is for single node is calculated. For implementation of proposed work is done with the help of banana species and month- wise requirement of water and entire cultivation requirement for one banana tree or node is calculated in the year of 2020 and 2021. With the help of experimentation of entire cultivation of 2020 the entire water requirements is 273 and optimized water is 127. The 31.14 % of water usage is reduced using the various parameters. Similarly, 2021 cultivation also around 50.1 percentage of water is optimized. Hence this work is achieved a novelty for water optimization and it could be used in many applications such as paddy, banana, sugercane, corrot cultivations in India and abroad with ranges of moisture and heat. Since, this work has introduced optimization of water usage for irrigation systems and certainly the novelty of efficient water usage proved. Thus the proposed work might be best suited in the irrigation system of hot land areas, step down agriculture of hills, and some rainy based agriculture cultivation systems. Finally when the external interference, and climate of location are being varies then optimization is also change considerably since difference locations have different moisture as key parameters. Also if noise of signal is considerably more then it also affect the water optimization. There would be direct and indirect correlation between performances of this proposed work with respect to climate changes in respective areas. The performance would be slightly more if the weather of land changes cold and performance would be slightly less if the area keeps hot than ordinary. It may be noticed that, is there any internal factor which reduces or increases the water efficiency for irrigation. The factors are sunny and rainy. The external factor rainy will be reducing the requirements of water in irrigation and another factor sunny will be increasing the requirements of water in irrigation. Usually wind will be changing the climate slightly to cool the environment, certainly increasing the efficiency of the irrigation system. In future work, ensemble and unsupervised learning algorithms are recommended to improve the accuracy of prediction and better decision making which could introduced deep prediction and might solve the issues of signal of noise and other external interfences. The water flow and land slop also considered for better prediction of requirement of irrigation system. In future, this work would be extended with introducing automation of irrigation and its algorithm would be coined. Also there is chance to introducing security of water pouring would be extended to strengthen this work. Auto irrigation systems will be implemented as future work and parameters considered here are moisture and humidity based rainy and sunny climate.

CRediT authorship contribution statement

D. Maria Manuel Vianny: Conceptualization, Methodology, Software, Data curation, Writing – original draft. **A. John:** Conceptualization, Methodology, Software. **Senthil Kumar Mohan:** Data curation, Writing – original draft. **Aliza Sarlan:** Validation and Reviewing final

draft. **Adimoolam:** . **Ali Ahmadian:** Supervision, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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